

Review retrospective:
the generic- (k_1, k_2) numerator lift — the prediction holds

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Abstract

A soundness review of `Laplace/TwoD/KthKthNumeratorAsymptotic.lean`: the generic- (k_1, k_2) asymptotic of the unnormalised 2D separable monomial numerator $\int x^{2j_1} y^{2j_2} e^{-tL}$, the second leg of the “2D lift” trio. Result: **a clean fidelity pass with one dead-code removal** — the dead `open Real` predicted by the partition-leg review, confirmed.

1 Setting

This is the numerator analogue of the partition leg reviewed immediately before it: same four-step lift pattern, same factorisation root (`AddSeparable`, here via `integral_separable_addSeparable`), same 1D base (`MonomialPotential`, here via `kth_moment_even`). The partition-leg retrospective predicted the same dead `open Real` would recur in the sibling files; this review confirms it.

2 What was reviewed

Three public theorems and the composite constant $C'(k_1, k_2, j_1, j_2) = \frac{1}{k_1 k_2} (2k_1)!^{(2j_1+1)/(2k_1)} (2k_2)!^{(2j_2+1)/(2k_2)} \Gamma(\frac{2j_1+1}{2k_1}) \Gamma(\frac{2j_2+1}{2k_2})$: the exact closed form C' . $t^{-(2j_1+1)/(2k_1) - (2j_2+1)/(2k_2)}$, the rescaled limit $\rightarrow C'$, and the asymptotic equivalence.

3 Fidelity / soundness findings

Sound; no mismatch. The composite constant is exactly the product of the two 1D `kth_moment_even` constants, and the exponent $-(2j_1+1)/(2k_1) - (2j_2+1)/(2k_2)$ is correct throughout (with the positive mirror in the rescaling). The hypotheses sit where they should ($t > 0$ for the exact form, only $k_i \geq 1$ for the `atTop` statements). Unlike the partition leg — whose module bullet had conflated two theorems and called a fractional power-law a “polynomial” — this file’s docstrings match their statements exactly; the reviewer found no over- or under-claim.

4 Simplifications applied

One dead open. Every `Real.exp/Real.Gamma/Real.div_rpow/Real.rpow.*` is qualified and the bare $t^{\wedge}(\text{real})$ is the `rpow` instance, so `open Real` is dead; `MeasureTheory` (bare $\int$), `Filter`, `Asymptotics`, and scoped `Topology` stay. Module green at 2700 jobs, no warnings, signatures byte-identical. GPT affirmed.

5 Disagreements and open questions

None. The reviewer’s other suggestions (replace the inline `show ... by ring` equality proofs inside `rw` with `change/named have`s; factor the twin `h_k1/h_k2` rewrites into a local helper) were declined as proof-style/structural refactors — and, as in the partition leg, `change` cannot replace an equality *proof term* passed to `rw`, so that one is not even warning-clearing.

6 Lessons

- **The sibling-prediction pattern holds again.** A dead-code finding in one instance of a copy-paste lift family is a near-certain finding in the rest — the dead `open Real` crossed cleanly from the partition leg to the numerator leg, as the `Quartic→Sextic` prediction first established. The cheap confirmation is one rebuild.
- **A clean docstring is the more common case.** The partition leg’s mislabelled bullet was the exception, not the rule; this sibling’s terse bullets were accurate. Catching the one drift does not mean the family is riddled with it.

7 Follow-ups

- The third leg, `KthKthMomentAsymptotic` (the Gibbs-moment lift, $= \text{numerator}/Z$), and `KthKthOddMoments`, plus the $(2, 3)$ `QuarticSextic*` specialisations — all near-certain repeats of the same dead `open Real`.
- Proof-golf forward tide: the `h_k1/h_k2` dedup helper, shared across the lift family.